

SBI 812

Machine Learning in Bioinformatics

Lecture 1: Introduction to Machine Learning and pandas

Central idea	Machine learning begins with a biological question and a trustworthy understanding of the data - not with a modelling function.
Practical theme	From a biological table to an analysis-ready dataset.
Main tools	pandas DataFrames, selection, filtering, missing values, summaries and reproducible inspection.

1. What is machine learning?

CORE IDEA

Machine learning learns statistical patterns from data in order to make predictions, assign labels, discover structure or simplify complex data.

CONVENTIONAL PROGRAMMING

data + human-written rules -> answer

SUPERVISED MACHINE LEARNING

labelled examples + learning algorithm -> fitted model

fitted model + new data -> predicted answer

MACHINE LEARNING DOES NOT AUTOMATICALLY

prove biological causation;

repair biased or poor-quality data;

understand what a biological column means;

guarantee performance in a different laboratory or population.

KEY POINT

Machine learning finds patterns. Biological interpretation and validation remain the researcher's responsibility.

WORKING SPACE

2. Four common machine-learning tasks

SUPERVISED LEARNING

Classification: target is categorical - resistant/sensitive, healthy/diseased.

Regression: target is continuous - growth rate, drug response, protein abundance.

UNSUPERVISED LEARNING

Clustering: identify groups of similar samples without supplied labels.

Dimensionality reduction: represent many measurements with fewer derived dimensions.

BIOINFORMATICS EXAMPLES

Tumour malignant/benign -> classification

Predict IC50 -> regression

Find unknown cell populations -> clustering

Represent 20,000 genes in two dimensions -> dimensionality reduction

IMPORTANT

A cluster is a mathematical grouping, not automatically a new biological subtype.

WORKING SPACE

3. The language of a machine-learning dataset

ONE ROW

Sample / observation: patient, tissue, cell, culture, sequence or other independent observation.

COLUMNS

Feature: input measurement used for learning or prediction.

Target: value a supervised model is trained to predict.

Label: categorical target.

Identifier: names a sample; normally not a predictive feature.

Metadata: describes sample or experiment.

sample_id | species | treatment | gene_a | gene_b |
growth_rate | response

COLUMN ROLES DEPEND ON THE QUESTION

If the question is 'Can gene expression predict response?', response is the target and gene measurements are features.

PREDICTION IS NOT EXPLANATION

A predictive feature may reflect correlation, batch, sampling strategy or another unrecorded factor rather than biological causation.

WORKING SPACE

4. pandas and the DataFrame

PANDAS

pandas is a Python library for working with labelled rectangular data.

```
import pandas as pd
```

READ THE IMPORT

import loads the library.

pandas is the library name.

as pd gives it the short alias pd.

```
expression_data =  
pd.read_csv("lecture01_bacterial_expression.csv")
```

READ FROM RIGHT TO LEFT

pd -> pandas

read_csv() -> read a comma-separated table

filename -> source file

expression_data = -> give the DataFrame a reusable name

```
expression_data.head(3)  
expression_data['gene_a']
```

WORKING SPACE

5. Inspect before analysing

FIRST RULE

Never assume that a file is correct merely because it loaded without an error.

```
expression_data.head()  
expression_data.tail(2)  
expression_data.shape  
expression_data.columns
```

```
expression_data.dtypes  
expression_data.info()  
expression_data.isna().sum()
```

CHECK

Expected number of rows and columns?

Are numerical measurements actually numeric?

Are identifiers stored as text?

Are values missing?

Are category spellings consistent?

```
expression_data.describe()  
expression_data['species'].value_counts()  
expression_data['species'].unique()
```

SUMMARY STATISTICS

Unusual minima or maxima are quality-control clues, not automatically biological discoveries.

WORKING SPACE

6. Selecting and filtering observations

SELECT ONE COLUMN

`expression_data['gene_a']` usually returns a Series.

SELECT SEVERAL COLUMNS

```
gene_columns = ['gene_a', 'gene_b', 'gene_c']
```

```
expression_data[gene_columns]
```

```
expression_data.loc[0:3,  
['sample_id', 'species', 'gene_a']]  
expression_data.iloc[0:4, 0:3]
```

LOC vs ILOC

loc works with labels.

iloc works with integer positions.

```
drug_samples = expression_data.loc[  
expression_data['treatment'] == 'drug'  
].copy()
```

```
low_growth = expression_data.loc[  
expression_data['growth_rate'] < 0.50,  
['sample_id', 'species', 'growth_rate']  
]
```

WORKING SPACE

7. Missing values and data types

MISSING DATA

A missing measurement may reflect failed assay, data entry, true absence of measurement or another technical/biological reason.

Deleting it automatically is not a scientific decision.

```
expression_data.loc[expression_data['gene_a'].isna()  
]  
expression_data['gene_a'].mean()
```

POSSIBLE RESPONSES

investigate the source;

repeat measurement if possible;

remove with justification;

impute appropriately;

use a method that handles missingness.

```
typed_data = expression_data.copy()  
typed_data['gene_a'] = pd.to_numeric(  
typed_data['gene_a'], errors='coerce'  
)
```

IMPORTANT

`errors='coerce'` converts unparseable values to missing values; compare missing-value counts before and after conversion.

WORKING SPACE

8. Summaries, grouped analysis and leakage

```
expression_data.groupby('treatment')['growth_rate']  
.mean()
```

```
growth_summary = (
    expression_data
    .groupby('treatment')['growth_rate']
    .agg(['count', 'mean', 'median', 'std'])
)
```

```
pd.crosstab(expression_data['treatment'],
expression_data['response'])
```

INTERPRET CAUTIOUSLY

A difference between group means does not by itself show causation.

Species composition, sample size, batch effects and measurement error may contribute.

DATA LEAKAGE

Leakage occurs when model development uses information that would not legitimately be available when predicting a new sample.

Identifiers, future measurements, duplicated/related samples and variables closely encoding the target can all create leakage.

WORKING SPACE

9. A reproducible inspection checklist

```
import pandas as pd

df =
pd.read_csv('lecture01_bacterial_expression.csv')
print('Shape:', df.shape)
print(df.columns.tolist())
print(df.dtypes)
print(df.isna().sum())
print(df.head())
print(df.describe())
```

THEN ASK

- What does one row represent?
- Which columns are identifiers, metadata, features and possible targets?
- Are units documented?
- Are categorical values consistent?
- Are measurements impossible or implausible?
- Are repeated measures or technical replicates present?
- Could related samples later appear in both training and test sets?

WORKING SPACE

10. AI-generated code must be audited

AI-GENERATED EXAMPLE

An AI assistant proposes:

```
df =  
pd.read_csv('lecture01_bacterial_expression.csv')  
df = df.dropna()  
X = df.drop(columns='response')  
print('The data are clean and ready for machine  
learning.')
```

PROBLEMS

`dropna()` removes rows without investigating missingness.

Removed samples are not reported.

`sample_id` remains among possible inputs.

Text columns still need appropriate handling.

The prediction question has not been defined.

Leakage, batch, related samples and implausible values were not checked.

'Clean' is asserted without evidence.

VERIFICATION HABIT

Explain each line -> test on a small known example ->
inspect the result -> compare with the biological question.

WORKING SPACE

11. Guided practical: what can we learn from this table?

INSPECT

How many samples and columns?

Which columns are numerical?

Which sample contains a missing value?

How many samples belong to each species?

How many are resistant?

SELECT AND FILTER

Select sample_id plus the three gene columns.

Select only control samples.

Select resistant B. subtilis samples.

Select samples with gene_b > 3.5.

Select the first five rows and first four columns with iloc.

SUMMARISE

Median of each gene.

Mean growth rate by species.

Mean growth rate by treatment.

Counts for each species-response combination.

REASON BIOLOGICALLY

Which columns could predict response?

Why should sample_id normally be excluded?

Could treatment leak information about response?

Does one missing value justify deleting the whole sample?

WORKING SPACE

12. End-of-lecture summary

MACHINE LEARNING STARTS BEFORE MODELLING

A precise biological question and trustworthy data come first.

DATASET LANGUAGE

sample -> one observation

feature -> model input

target -> value to predict

identifier -> sample name

metadata -> contextual information

PANDAS WORKFLOW

Load -> Inspect -> Select -> Filter -> Check missingness/types -> Summarise -> Save

```
pd.read_csv()  
head() shape dtypes isna()  
loc iloc  
groupby() value_counts()  
to_csv()
```

FINAL QUESTION

What evidence supports the claim that a biological table is ready for machine learning?

WORKING SPACE